Week 2
Course: Introduction to programming 2 Theme: Conditional statement, logical operators, while loop, for loop, strings

Part 1. Theoretical Exercise

1. c. decision

2. b. single alternative decision

3. d. Boolean

4. a. relational

5. c. dual alternative decision

6. b. if

7. c. if-else

8. b. logical

9. a. and

10. b. or

11. c. not

12. a. flag

1. True

2. False

3. False

4. False

5. True

6. True

1. Conditionally executed means that code runs only when a specific condition (a logical expression) is True.

2. Dual alternative decision structure (if-else statement).

3. The **and** operator combines two expressions. The result is True only if both expressions are True; otherwise, it is False.

4. The **or** operator combines two expressions. The result is True if at least one expression is True; it is False only if both are False.

5. To check if a number is within a range (e.g., from 10 to 20), the **and** operator is used (e.g., x >= 10 and x <= 20).

6. A flag is a variable (usually of type bool) that indicates a state or whether a condition has been met in a program.

Problem 1

```python
def letter():  1 usage  new *
    letter = str(input("Enter the letter: "))
    #a, e, i, o, u
    if letter == "a" or letter == "e" or letter == "i" or letter == "o" or letter == "u":
        print("letter is vowel")
    elif letter == "y":
        print("y sometimes vowel , sometimes consonant")
    else:
        print("letter is consonant")
letter()
```

Run problem1

```
/Users/nurislamaldabergenuly/PycharmProjects/PythonProject/.venv1/bin/python /Users/nurislamaldabergenu
Enter the letter: a
letter is vowel
```

Problem 2:

```python
def identify_shape():  1 usage  new *
    try:
        sides = int(input("Введите количество сторон фигуры (от 3 до 10): "))


        if sides == 3:
```

problem2

```
/Users/nurislamaldabergenuly/PycharmProjects/PythonProject/.venv1/bin/python /Use
Введите количество сторон фигуры (от 3 до 10): 5
Это пятиугольник
```

Problem 3

```python
def days_in_month():  1 usage  new *
    else:
        print("Ошибка неверное название месяца.")


days_in_month()
```

Run problem3

```
/Users/nurislamaldabergenuly/PycharmProjects/PythonProject/
Введите название месяца: January
В месяце January 31 день.
```

Problem 4

```python
def problem4():  1 usage  new *
        season = "Spring" if day < 21 else "Summer"
    elif month == "July" or month == "August":
        season = "Summer"
    elif month == "September":
        season = "Summer" if day < 22 else "Fall"
```

Run problem4

```
/Users/nurislamaldabergenuly/PycharmProjects/PythonProject/.venv1/bin
Enter the name of the month: January
Enter the day: 2
Winter
```

Problem 5

```python
def problem5():  1 usage  new *
    grade = input().strip().upper()

    if grade == "A+" or grade == "A":
        points = 4.0
    elif grade == "A-":
```

Run problem5

```
/Users/nurislamaldabergenuly/PycharmProje
A+
4.0
```

Problem 6

```python
def problem6():  1 usage  new *
    total = 0
    count = 0

```

Run problem6

```
/Users/nurislamaldabergenuly/PycharmProjects/PythonProject/.venv1/bin/python /Users/nurislam
37
7
8
8
0
15.0
```

Problem 7

```python
def problem7():  1 usage  new *
    print(f"{'Celsius':<10} | {'Fahrenheit':<10}")
    print("-" * 25)
```

Run problem7

```
/Users/nurislamaldabergenuly/PycharmProjects/PythonProject/.venv1/bin/python /Users/nu
Celsius    | Fahrenheit
-------------------------
0          | 32.0
10         | 50.0
```

Problem 8

```python
def problem8():  1 usage  new *
    total = 0

    while True:
```

Run problem8

```
/Users/nurislamaldabergenuly/PycharmProjects/PythonProject/.venv1/bin/python /Users/nu
18
33
6

60.00
```

Problem 9

```python
def problem9():  1 usage  new *
    total_cost = 0.0

```

Run python9

```
/Users/nurislamaldabergenuly/PycharmProjects/PythonProject/.venv1/bin/python /Users/nurislama
1
2
3

The total admission cost for the group is 14.00
```

Problem 10

```python
if __name__ == "__main__":
    problem10()
```

Run python10

```
/Users/nurislamaldabergenuly/PycharmProjects/PythonProject/.venv1/bin/python /Users/nuri
Enter a number: 1
1.0 x 1 = 1.0
1.0 x 2 = 2.0
1.0 x 3 = 3.0
1.0 x 4 = 4.0
```